
EC9600: Applied Algorithms

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Code	EC9600
Title	Applied Algorithms
Academic Credits	02
Prerequisite/s	EC4010
LMS (Teams)	E22_EC9600_team
Coordinator	Dr. U. Thanuja
Lecturers	Dr. U. Thanuja

Learning Outcomes

By the end of this course unit, students should be able to:

- ILO1: Describe real world applications of algorithms;
- ILO2: Utilize combinatorial algorithms for solution space search;
- ILO3: Perform exploratory analysis on huge and rapidly changing data;
- ILO4: Explain game theory as a mechanisms to achieve efficient and desirable global outcomes in spite of the selfish behaviour;
- ILO5: Comprehend the use and the differences between localized algorithms as opposed to centralized algorithms.

Course Contents	Hours			
	L	T	L/F	A
1. Overview Revision on algorithms and their time and space complexities; Examples of applications of algorithms in real world.	03			
2. Combinatorial Algorithms and Graph Theory Generating all and random instances of a combinatorial object in molecular; Exhaustive search through the solution space, which are represented as combinatorial structures such as permutations, combinations, set partitions, integer partitions, and trees; Graph theoretic models in molecular biology: RNA, proteins, and other structures described as graphs	04			03
3. Bioinformatics Algorithms Methods for the analysis of gene expression data.	04			06
4. Processing Data Streams Exploratory analyses of huge and rapidly changing data streams such as network traffic, online auctions, transaction logs, telephone call records, automated bank machine operations, and atmospheric and astronomical events.	04			06
5. Game Theory Game theory and applications; Nash equilibrium; algorithmic solutions and advances achieved through game theory	04			03

Course Contents	Hours			
	L	T	L/F	A
6. Localized Algorithms Topology control for wireless ad-hoc or sensor networks; Neighbour elimination schemes, which remove edges from the initial connection graph in order to generate energy efficient, sparse, planar but still connected network in localized manner.	04			03
Total Hours = 23 + 21/3 = 30	23			21

Course Assessment / Evaluation

Assessment Type	Assessment Method	Covered ILOs	Group / Individual	Percentage (%)
In-course Assessment	Assignments	1, 2, 3, 4, 5	Group (02)	20
	Quiz	2, 3, 4, 5	Individual (04)	20
	Mid Semester Assessment	All	Individual	20
End-of-course evaluation	End Semester Examination	All	Individual	40

Course Assessment Schedule

		Week No														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Assessment Methods	Programming Assignment - 01					X										
	Programming Assignment - 02										X					
	Quiz - 1			X												
	Quiz - 2						X									
	Quiz - 3									X						
	Quiz - 4												X			
	MID								X							

Reference books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms, fourth edition*. MIT Press, 2022. (Not recommended in syllabus)

Class Rules

- Be punctual to lectures and quizzes. You are going to be Engineers.
- 80% attendance to lectures (common rule).
- Should not use mobile phones or laptops unless asked.
- Please submit assignments and quizzes on time.
- To pass this course, you should get ≥ 35 for both In-course assessment and End semester exam.

Questions?